Taxonomy of Medical Concepts Extractable from Clinical Notes

Executive Summary

Clinical notes contain a rich, but messy, set of medical concepts that can be systematically extracted and normalized. These concepts fall into a relatively stable set of categories used across research, open-source toolkits, and commercial clinical NLP systems.

- The core extractable concepts cluster into about **20–30 entity types**, organized under a few macro-categories: **conditions & clinical states, diagnostics & measurements, therapeutics & interventions, patient context & history, oncology-specific entities, care planning & outcomes, and risk & determinants**.

- For each entity (e.g., “NSTEMI”, “metoprolol 25 mg PO BID”, “HbA1c 9.2%”), there are critical **contextual attributes**—negation, temporality, certainty, experiencer, severity, anatomical site, stage, dosage, etc.—that must be extracted alongside the base concept to make the data clinically safe and analytically useful.

- These extracted concepts align cleanly with **standard terminologies and data models**: disorders and findings → SNOMED CT / ICD-10 and FHIR `Condition`; labs and vitals → LOINC and FHIR `Observation`; procedures → SNOMED CT / CPT / ICD-10-PCS and FHIR `Procedure`/`ServiceRequest`; medications → RxNorm and FHIR `Medication*` resources; social and family history → FHIR `Observation` / `FamilyMemberHistory`.

- Different note types (H&P, progress, discharge, radiology, pathology, operative, nursing) emphasize **different subsets of these concepts** and provide complementary views of the patient journey. A robust extraction strategy understands where each concept “naturally lives” in the documentation.

- When designing clinical NLP on a platform like Snowflake, it is useful to think of text-extracted items as **standard-coded FHIR-like entities plus attributes**, rather than as ad-hoc strings. This enables interoperability, longitudinal analytics, and model reuse across engines and sites.

Detailed Analysis

1. Where these concepts come from: note types and sections

Clinical concepts appear across multiple note types and structured sections. Knowing where they typically appear helps design extraction rules, QA, and downstream models.

Common note types

- **History & Physical (H&P) / Admission note**
  Rich in: chief complaint, history of present illness, past medical/surgical history, medications, allergies, social & family history, review of systems, initial exam, initial assessment & plan.

- **Progress notes / daily notes / consult notes**
  Emphasize: interval history, response to treatment, new problems, evolving diagnoses, updated labs and imaging, and updated plan.

- **Discharge summaries / transfer summaries**
  Dense summary of: final diagnoses, hospital course, key procedures, complications, important test results, discharge medications, discharge condition, follow-up plans.

- **Radiology reports** (CT, MRI, X-ray, ultrasound, nuclear medicine)
  Highly structured narrative: indication, technique, findings, impression → great source for imaging findings, tumor measurements, staging clues.

- **Pathology & cytology reports**
  Contain: specimen descriptions, histology, grade, margins, molecular markers → critical for oncology concepts (tumor type, grade, receptor status).

- **Operative / procedure notes**
  Detail: procedures performed, indications, techniques, devices/implants used, intraoperative findings, complications.

- **Nursing notes / flowsheets**
  Emphasize: vitals, intake/output, pain scores, functional status, bedside observations, nursing assessments, SDoH clues.

- **Emergency department notes, triage notes**
  Emphasize: acute complaints, initial vitals, quick problem list, risk scores.

Common sections within physician notes

- **Chief Complaint (CC)** – symptom or reason for encounter: “Chest pain”, “Shortness of breath”.
- **History of Present Illness (HPI)** – timeline, symptoms, exacerbating/relieving factors.
- **Past Medical History / Past Surgical History (PMH/PSH)** – chronic conditions, prior surgeries.
- **Medications** – home meds, inpatient meds.
- **Allergies** – drug/food/environmental allergies and reactions.
- **Social History** – tobacco, alcohol, drugs, occupation, living situation.
- **Family History** – diseases in relatives and ages of onset.
- **Review of Systems (ROS)** – systematic positive/negative symptoms by organ system.
- **Physical Exam (PE)** – objective findings by organ system.
- **Labs / Studies** – lab panels, imaging, ECG, other tests.
- **Assessment & Plan (A/P)** – problem-oriented diagnoses and plans; high value for structured extraction.

Each of the concept categories below tends to be concentrated in specific note sections (e.g., signs/symptoms in CC/HPI/ROS; procedures in operative notes; staging in pathology/radiology reports), which you can leverage for higher-precision extraction and validation.

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2. High-level categories of extractable medical concepts

This section lays out the main categories and key subtypes, with definitions, note-style examples, and common standards for normalization.

2.1 Conditions & clinical states

These are the backbone of almost any clinical NLP system.

2.1.1 Problems / diagnoses

Definition: Clinically recognized diseases, disorders, or diagnoses that require or have required management.

Examples (as written in notes)

- “Admitted with **NSTEMI**.”
- “Active problems: **COPD**, **HTN**, **hyperlipidemia**.”
- “History of **ischemic stroke** in 2019.”
- “New diagnosis of **congestive heart failure**.”

Standards / mappings

- **SNOMED CT**: Clinical finding / Disorder hierarchy
- **ICD-10-CM**: Diagnosis codes
- **MedDRA**: For adverse event reporting
- **UMLS**: *Disease or Syndrome*, *Pathologic Function*
- **FHIR**: `Condition` (with clinicalStatus, verificationStatus, category)

2.1.2 Signs & symptoms

Definition: Subjective complaints (symptoms) and objective abnormal signs not necessarily formulated as formal diagnoses.

Examples

- “Complains of **shortness of breath** on exertion.”
- “Denies **chest pain** or **palpitations**.”
- “Reports **fatigue**, **night sweats**, and **weight loss**.”
- “Exam: **S3 gallop** present, **bilateral wheezes**.”

Standards

- **SNOMED CT**: Clinical finding (symptom/sign descendants)
- **ICD-10-CM**: R-codes (symptoms and abnormal findings)
- **FHIR**: Often `Condition` or `Observation` depending on modeling
- **UMLS**: *Sign or Symptom*

2.1.3 Clinical findings (exam, imaging, test interpretations)

Definition: Objective findings from physical exam or interpretation of tests.

Examples

- “CXR: **right lower lobe consolidation**.”
- “Echo shows **EF 25%**, **global hypokinesis**.”
- “PE: **+1 pitting edema** in both legs.”
- “ECG with **atrial fibrillation**.”

Standards

- **SNOMED CT**: Clinical finding, Morphologic abnormality
- **LOINC**: For structured measurements (ejection fraction, lab values)
- **FHIR**: `Observation` (with interpretation, bodySite, method)

2.1.4 Comorbidities & chronic conditions

Often the same concept type as diagnoses, but of special interest for risk and quality measures.

Examples

- “Long-standing **type 2 diabetes mellitus**, poorly controlled.”
- “**Stage 3 CKD** due to diabetic nephropathy.”
- “**Rheumatoid arthritis** on biologic therapy.”

Typically represented as `Condition` resources tagged as chronic and long-standing.

2.1.5 Functional status & performance scales

Definition: Functional capacity and performance, important especially in geriatrics and oncology.

Examples

- “**ECOG 2** – ambulatory but unable to work.”
- “Requires assistance with **ADLs** (bathing, dressing).”
- “**NYHA class III** heart failure symptoms.”

Standards

- **SNOMED CT** and **NCIt**: Performance status, NYHA, ADL concepts
- **FHIR**: `Observation` with coded scales and numeric/ordinal values

2.2 Diagnostics & measurements

2.2.1 Laboratory tests (orders & results)

Definition: Lab tests and their results (numeric or categorical).

Examples

- “Labs: **Na 128**, **K 5.6**, **Cr 2.1**.”
- “Order **CBC with diff**, **troponin q6h ×3**.”
- “**HbA1c 9.2%** (poorly controlled).”

Standards

- **LOINC**: Test identifiers
- **UCUM**: Units (mg/dL, mmol/L, %)
- **FHIR**: `Observation` (results), `ServiceRequest` (orders)

2.2.2 Imaging & other diagnostic tests

Definition: Imaging and physiologic tests as both procedures and findings.

Examples

- "**CT chest with contrast**: spiculated **2.5 cm LUL mass**."
- "**MRI brain**: no acute infarct."
- "**Pulmonary function tests** consistent with moderate obstruction."

Standards

- **LOINC**: Imaging test names
- **SNOMED CT**: Imaging procedures and findings
- **FHIR**: `ImagingStudy`, `DiagnosticReport`, `Observation`, `Procedure`

2.2.3 Vitals & physiologic measurements

Definition: Routine bedside measurements.

Examples

- "VS: **BP 160/90**, **HR 110**, **RR 24**, **T 38.5°C**, **SpO2 89%** on RA."
- "**Weight 95 kg**, **BMI 33**."

Standards

- **LOINC**: Vital sign codes
- **FHIR**: `Observation` (vital signs profile)

2.2.4 Clinical scores & risk calculators

Definition: Scores like CHA_2DS_2-VASc, APGAR, Wells, SOFA, etc.

Examples

- "**CHA_2DS_2-VASc score 4** (moderate stroke risk)."
- "**SOFA 9** on admission."

Standards

- **LOINC** / **SNOMED CT** / **NCIt**: Many scores have codes
- **FHIR**: `Observation` (value is the score)

2.3 Therapeutics & interventions

2.3.1 Procedures (surgical, diagnostic, therapeutic)

Definition: Actions performed for diagnosis or treatment.

Examples

- “Underwent **laparoscopic cholecystectomy** on 3/14.”
- “Planned **colonoscopy** next week.”
- “Status post **CABG ×3** in 2016.”

Standards

- **SNOMED CT**: Procedures
- **ICD-10-PCS**, **CPT/HCPCS**: Billing and procedure codes
- **FHIR**: `Procedure` (performed), `ServiceRequest` (ordered)

2.3.2 Medications (orders, prescriptions, administrations)

Definition: Drug products and their use.

Examples

- “Started on **metoprolol tartrate 25 mg PO BID**.”
- “Home meds: **Lantus 30 units qHS**, **Lipitor 40 mg daily**.”
- “Hold **warfarin** 5 days prior to surgery.”

Key attributes (see cross-cutting section later): strength, dose, route, frequency, duration, indication.

Standards

- **RxNorm**: Ingredient and product codes
- **SNOMED CT**: Pharmaceutical/biologic products
- **FHIR**: `Medication`, `MedicationRequest`, `MedicationAdministration`, `MedicationStatement`

2.3.3 Devices & implants

Definition: Medical devices in or on the patient.

Examples

- "Has **dual-chamber pacemaker** implanted 2015."
- "On **LVAD** awaiting transplant."
- "**Right hip prosthesis** in situ."

Standards

- **SNOMED CT**: Medical devices
- **GMDN/UMDNS**: Device nomenclature (outside text, but target codes)
- **FHIR**: `Device`, `DeviceUseStatement`, `Procedure` (for placement)

2.3.4 Radiation therapy & chemotherapy regimens

Definition: Cancer therapies as procedures and regimens.

Examples

- "Completed **4 cycles of FOLFOX**."
- "Receiving **IMRT 60 Gy in 30 fractions** to mediastinum."

Standards

- **SNOMED CT / NCIt**: Chemo agents and regimens, radiation modalities
- **FHIR**: `MedicationAdministration`/`MedicationRequest` for chemo; `Procedure` for radiotherapy

2.4 Patient context & history

2.4.1 Demographics

Definition: Age, sex/gender, language, marital status, etc.

Examples

- "**67-year-old male** with history of…"
- "**Spanish-speaking**, requires interpreter."

Standards

- **FHIR**: `Patient` (birthDate, gender, communication, etc.)

2.4.2 Social history & social determinants

Definition: Behavioral, lifestyle, socioeconomic, environmental context.

Examples

- "**Smokes 1 ppd × 30 yrs**, not ready to quit."
- "Drinks **2–3 beers nightly**."
- "Lives alone; limited **social support**."
- "History of **IV drug use**; in recovery."

Standards

- **SNOMED CT**: Tobacco, alcohol, drug use, housing, etc.
- **ICD-10-CM**: Z-codes for SDoH (e.g., housing, employment)
- **FHIR**: `Observation` (e.g., smoking status), `QuestionnaireResponse`, SDoH profiles

2.4.3 Family history

Definition: Conditions in family members that affect risk.

Examples

- "FHx: **mother – breast cancer at 45**; **father – MI at 52**."
- "No family history of **colon cancer** or **diabetes**."

Standards

- **ICD-10-CM**: Family history Z-codes
- **SNOMED CT**: Family history situations
- **FHIR**: `FamilyMemberHistory` + nested `Condition`s

2.4.4 Past medical, surgical, obstetric history

Definition: Historical diagnoses, surgeries, and pregnancies.

Examples

- "PMH: **HTN**, **T2DM**, **prior DVT**."
- "PSH: **appendectomy**, **C-section ×2**."

Modeled largely as `Condition` and `Procedure` with temporal context = past.

2.4.5 Allergies & intolerances

Definition: Hypersensitivity or other clinically significant reactions.

Examples

- “Allergies: **penicillin – anaphylaxis**, **sulfa – rash**.”
- “No known drug allergies (**NKDA**).”

Standards

- **SNOMED CT**: Allergic disposition, causative agents
- **RxNorm**: Allergen drugs
- **MedDRA**: Reaction terms
- **FHIR**: `AllergyIntolerance` (with substance and manifestations)

2.4.6 Adverse events (including ADEs)

Definition: Unintended harmful events related to drugs, devices, or procedures.

Examples

- “Developed **GI bleed** likely from **NSAID** use.”
- “Post-op **wound infection**.”
- “Contrast-induced **AKI**.”

Standards

- **MedDRA**: Primary for adverse events
- **SNOMED CT / ICD-10-CM**: Disorders/complications
- **FHIR**: `AdverseEvent`, `Condition`, `Observation`

2.5 Oncology-specific concepts

Oncology is often a special focus of clinical NLP; notes and pathology/radiology reports expose several specific concept families.

2.5.1 Tumor diagnosis and site

Examples

- “**Invasive ductal carcinoma** of the left breast.”

- “**Stage IIIB (T3N2M0) NSCLC**, squamous histology.”

Standards

- **SNOMED CT**: Neoplastic processes
- **ICD-10-CM**: Cancer diagnoses
- **ICD-O-3**, **NCIt**: Site + morphology
- **FHIR**: `Condition` (primary cancer, metastatic sites)

2.5.2 Histology, grade, biomarkers

Examples

- “Grade **3** tumor, **Gleason 4+3=7**.”
- “**ER+/PR+/HER2–** breast cancer.”
- “Tumor is **KRAS G12D mutant**.”

Standards

- **SNOMED CT / NCIt**: Histologic types, grades, biomarkers
- **FHIR**: `Observation` (for receptor status, mutation calls), `MolecularSequence`

2.5.3 Stage & response

Examples

- “**Stage IVB** Hodgkin lymphoma.”
- “After 4 cycles, **partial response** by RECIST.”

Standards

- **SNOMED CT** and **NCIt**: TNM, stage group, response categories
- **FHIR**: `Condition.stage`, `Observation` (staging, response)

2.6 Care planning & management

2.6.1 Care plans, orders, and recommendations

Examples

- “Plan: **start ACE inhibitor**, **schedule stress test**, **low-salt diet**.”
- “Follow up with **cardiology in 2 weeks**.”
- “Refer to **pulmonary rehab**.”

Standards

- **SNOMED CT**: Planned procedures, regimens
- **FHIR**: `CarePlan`, `ServiceRequest`, `MedicationRequest`, `Goal`

2.6.2 Outcomes & disposition

Examples

- "Patient **improved**, discharged home."
- "Expired on **hospital day 7** due to septic shock."
- "Course complicated by **pneumonia** and **prolonged ventilation**."

Standards

- **SNOMED CT**: Outcome concepts
- **ICD-10-CM**: Complications, cause of death
- **FHIR**: `Encounter` (dischargeDisposition), `Condition` (clinicalStatus), `Observation` (functional outcomes)

2.7 Risk & determinants

2.7.1 Clinical and genetic risk factors

Examples

- "**Obesity (BMI 38)**, **smoker**, **family history of premature CAD**."
- "**BRCA1 mutation positive**."
- "Poor **medication adherence**."

Standards

- **SNOMED CT**: Risk states, obesity, tobacco use, etc.
- **ICD-10-CM**: Z-codes for risk and personal history
- **FHIR**: `Observation` (risk scores, genetic variants), `Condition` (risk conditions)

2.7.2 Contraindications & precautions

Examples

- "Not a candidate for anticoagulation due to **recurrent GI bleed**."
- "Avoid **beta-blockers** given history of **severe asthma**."

Typically modeled as relationships: medication/procedure ↔ contraindicating condition.

2.8 Administrative & encounter context

Less obviously “clinical”, but often required for analysis and data modeling.

Examples

- Encounter type (ED visit, inpatient, outpatient).
- Location (ICU vs ward).
- Provider roles (attending, consultant).
- Service (cardiology, oncology).

Generally derived from structured EHR fields; sometimes confirmed or corrected in text (“Transferred from OSH”, “Seen in oncology clinic”).

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3. Cross-cutting contextual attributes

Extracting just the entity mentions is not enough. To be clinically faithful and analytically useful, you need to capture the **context** around each concept. Most mature clinical IE systems attach attributes like these to each entity:

3.1 Negation / affirmation

- Is the condition or symptom **present**, **absent**, or **unknown**?

Examples

- “**Denies chest pain**.” → chest pain: negated
- “No evidence of **pneumonia**.” → pneumonia: negated
- “Cannot exclude **PE**.” → PE: uncertain (not simply positive or negative)

Implementation pattern: an attribute `polarity ∈ {positive, negated}` plus an uncertainty dimension.

3.2 Temporality

- Is it **historical**, **current**, **future/planned**, or **resolved**?

Examples

- "**History of MI in 2018**." → MI: past
- "Will **schedule colonoscopy**." → colonoscopy: future/planned
- "**Resolved DKA**." → DKA: resolved, past

Map to Condition `clinicalStatus` and `onset/abatement` or Procedure/ServiceRequest `status`.

3.3 Certainty / assertion strength

- Is the diagnosis **definite**, **probable**, **possible**, or **ruled out**?

Examples

- "**Probable pneumonia**." → pneumonia: probable
- "Admit to **r/o PE**." → PE: under evaluation, not confirmed
- "Unlikely to represent **malignancy**." → malignancy: unlikely

Often an attribute `certainty` or encoded via `verificationStatus` in FHIR (provisional, differential, confirmed, refuted).

3.4 Experiencer

- Does the statement apply to the **patient** or someone else (usually a **family member**)?

Examples

- "**Mother with breast cancer** at 45." → breast cancer: experiencer = family
- "Brother with **bipolar disorder**." → bipolar disorder: family

Used to decide between `Condition` on the patient vs `FamilyMemberHistory`.

3.5 Severity

- Degree of intensity: **mild**, **moderate**, **severe**, **life-threatening**, etc.

Examples

- "**Severe COPD** with frequent exacerbations."
- "**Mild anemia**."

Models as a `severity` attribute on Condition, often mapped to SNOMED CT severity codes.

3.6 Anatomical site & laterality

- Body location and side: **left**, **right**, **bilateral**, etc.

Examples

- "**Left MCA stroke**."
- "**Right hip fracture**."
- "**Bilateral pleural effusions**."

Implement using a separate `AnatomicalSite` entity (SNOMED CT body structure) and/or `bodySite` attribute, plus `laterality`.

3.7 Medication-specific attributes

For medication entities, key attributes include:

- **Strength / dose** – "25 mg", "30 units", "2 mg/kg".
- **Route** – "PO", "IV", "IM", "subQ", "inhaled".
- **Frequency** – "BID", "q6h", "qHS", "PRN".
- **Duration** – "for 7 days", "x10 doses".
- **Indication** – which condition it is treating ("for **AFib**").

These align closely with FHIR `MedicationRequest.dosageInstruction`.

3.8 Stage, grade, and performance status

Critical in oncology and chronic disease.

- **Stage** – "Stage IIIB (T3N2M0)", "Stage IV".
- **Grade** – "grade 3", "Gleason 4+3=7".
- **Response** – "complete response", "stable disease", "progression".
- **Performance** – "ECOG 2", "Karnofsky 70%".

Represented as Observations linked to the cancer Condition, or as `Condition.stage` components.

3.9 Course / acuity

- Acute vs chronic, improving vs worsening, recurrent.

Examples

- "**Acute on chronic** heart failure."
- "**Recurrent** UTIs."
- "Symptoms **improving** on current therapy."

This can be encoded as attributes or separate course-modifying observations.

3.10 Hypothetical / conditional / planned

- Distinguish **plans**, **instructions**, and **hypotheticals** from documented facts.

Examples

- "If fever persists, **start antibiotics**." → future conditional.
- "Candidate **for transplant**." → not yet performed.

Useful to avoid over-counting procedures or medications that are merely considered.

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4. Aligning extracted concepts with standard models

For an architect or platform designer, it is helpful to think in three aligned layers:

1. **Semantic grouping**: UMLS semantic types or your own "entity classes" (Disease, Symptom, Medication, Lab, Procedure, Anatomy, Device, Social History, etc.).
2. **Standard terminologies**: SNOMED CT, ICD-10, RxNorm, LOINC, MedDRA, NCIt.
3. **Data model / exchange**: FHIR resources (`Condition`, `Observation`, `Procedure`, `Medication*`, `CarePlan`, etc.), or comparable canonical models.

A pragmatic pattern:

- Use **entity types** (e.g., `MEDICAL_CONDITION`, `MEDICATION`, `TEST_TREATMENT_PROCEDURE`, `ANATOMY`) that are **stable across NLP engines**.
- Normalize each entity to one or more **standard codes** (SNOMED CT/ICD for conditions, RxNorm for meds, LOINC for labs, etc.).
- Persist normalized entities and their **attributes** in a FHIR-like schema (or an internal canonical model) in your data platform.
- Preserve links to **source text spans and document/section context** for auditability and error analysis.

Conclusion

Clinical notes encode a broad but well-structured universe of medical concepts. When viewed through the lens of modern clinical NLP and standards, these concepts can be organized into:

- A manageable set of **core entity categories** (conditions, diagnostics, interventions, context, oncology, care plans, outcomes, risk) with well-defined subtypes.

- A critical layer of **contextual attributes** (negation, temporality, certainty, experiencer, severity, anatomy, dosage, stage, performance) that determine clinical meaning.
- A normalization layer that ties everything into **standard terminologies and data models** (SNOMED CT, ICD-10, RxNorm, LOINC, FHIR).

For your purposes—thinking about "what can we extract from clinical notes"—the key takeaway is that you can expect to reliably extract:

- **What is wrong** with the patient (diagnoses, findings, symptoms, comorbidities, risk factors).
- **What was measured** (labs, vitals, imaging and their interpretations, clinical scores).
- **What was done or planned** (procedures, medications, care plans, referrals).
- **Who the patient is and their context** (demographics, social and family history, allergies, adverse events).
- **How the disease evolves and ends** (course, stage, response, outcomes, disposition).

If you'd like next, I can:

- Propose a concrete **concept schema** (tables or FHIR-like resources) you could implement on Snowflake, or
- Prioritize which concept categories to target in a first clinical NLP deployment for, say, cardiology or oncology use cases.